

Phase 0 — Clear the blockers

Where Phase 0 stands, what Phase 1 inherits, and what a full audit of the work found. Built 21 September 2026, 13:47.

Phase 0 is 11 of 17 tickets done. Every ticket assigned to Atharva is finished and committed — the units problem, the bearing catalogue, the acceleration-to-velocity path, the tenfold cost cut and the test harness. Six are open, all of them Laxman's: the repository is not pushed, the two ingest paths still duplicate the pipeline, the upload write is not re-runnable, and there is no background worker.

The finding that changes what the system can say today: the gateway knows the readings are in g and knows each channel's sensitivity, and sends neither to the platform. So the database still records the unit as unconfirmed, and the safety rule built in VIK-005 refuses to grade any of the 94 captures. The work is done at both ends and not joined in the middle.

The finding that changes what it can say tomorrow: at the capture length the gateway is set to, three of the four bearing frequencies on this pump cannot be told apart from ordinary shaft harmonics. That is not a software limit and no amount of analysis fixes it. One setting on the Senvia sensor page does.

What Phase 0 was for

Phase 0 is not new features. It is the five things the roadmap said would make everything built on top of them wrong, plus somewhere to put tests. Each one is a correctness problem rather than a missing piece — the system already produced numbers for all five. The numbers were just not trustworthy.

- 1

The code lived on one laptop

Not a coding task. A repository now exists, holding the domain data files that a single wrong line in .gitignore had been silently excluding.
- 2

The chatbot's reference data was missing from disk

bearings.json, iso10816_3.json and fault_signatures.json are tracked. Every bearing, ISO and fault lookup used to raise an error on first call.
- 3

The system did not know what its own numbers meant

Readings were stored exactly as they arrived, with no record of whether they were volts or g — a factor of ten apart. There is now a column for it, and a rule that refuses to grade an undeclared capture instead of guessing.
- 4

The bearing catalogue sat unused in a spreadsheet

88,734 bearings are loaded. The pump's 6312-C3 and 6310-C3 now resolve to real catalogue entries instead of an estimate that could be wrong by one ball.
- 5

All the heavy maths ran while the user waited

Restructured: one extraction per segment instead of one per feature per segment. A whole 8-channel capture now takes 0.18 s, carrying 36 features instead of 10.

Ticket by ticket

Status was checked against the code and the database, not recalled. The evidence column says what was looked at.

ID	Task	Owner	Status	Evidence
VIK-001	Put the working copy under git	Laxman	Partly	Repository exists. 40 commits sit unpushed on this laptop; the remote branch was last updated on 2026-09-19.

ID	Task	Owner	Status	Evidence
VIK-002	Restore the chatbot's domain data	Atharva	Done	All three JSON files tracked in git.
VIK-003	Declare jinja2	Atharva	Done	Pinned at 3.1.3 in requirements.
VIK-004	Port units.py into the backend	Atharva	Done	backend/app/ai/units.py, with a test that fails if it drifts from the original.
VIK-005	Apply sensor sensitivity in the measurement path	Atharva	Partly	The code is written and tested. The value never arrives — see below.
VIK-006	Migration: signal_unit and unit_confirmed	Atharva	Done	Migration 022, applied, with a working downgrade. Renumbered from 019 after a collision with the bearing migration.
VIK-007	Acceleration to velocity	Atharva	Done	Reuses the existing integrator; no second one was added.
VIK-008	Migration: bearings table	Atharva	Done	Migration 019, applied.
VIK-009	Import the bearing spreadsheet	Atharva	Done	88,734 rows loaded. Safe to re-run.
VIK-010	Resolve equipment bearings against the catalogue	Atharva	Done	6312-C3 and 6310-C3 both resolve. The readiness flag is computed from the catalogue id, so it is no longer a box anyone can tick.
VIK-011	Route both ingest paths through run_pipeline	Laxman	Open	run_pipeline exists but no router calls it. Both paths still inline the sequence.
VIK-012	Make save_upload_data re-runnable	Laxman	Open	Still a plain INSERT. A second run on the same upload fails.
VIK-013	processing_jobs table and worker	Laxman	Open	No migration and no table. The latest migration is 022.
VIK-014	Cut segment-trend work tenfold	Atharva	Done	Loops inverted. 33 extraction calls where there were 330.
VIK-015	pytest and the first test module	Laxman	Partly	269 backend tests pass. pytest and httpx are now declared and collection is scoped to tests/, so the suite runs from a fresh checkout. Converting and deleting scripts/test_auth_phase1.py is still outstanding.
VIK-016	vitest and the first frontend test	Laxman	Partly	vitest runs and 43 tests pass. The ticket named lib/vibration-features.ts as the first thing to cover, and it is not covered.
VIK-017	Synthetic fault harness with recorded answers	Atharva	Done	Eight signatures, each with its expected answer stored alongside and checked against the waveform.

The gap that matters: the unit is known, and never arrives

VIK-005 built the rule that a capture with no declared unit is refused rather than graded. That rule is working. What it is refusing is every capture we have, and it should not be.

The gateway knows

SAMPLE_UNIT=g and SENSITIVITY_MV_PER_G=500,500,100,100,100,100,100,100 are set and correct. Channels 1 and 2 are 500 mV/g sensors; the cloud configuration's single value of 100 is wrong for them.

OK

my_script.py attaches it	sample_metadata() puts sampleUnit and sensitivityMvPerG on the upload, and flags that the sensitivities are not uniform.	OK
but only on the Senvia leg	platform_push.py, which is the leg feeding our own platform, sends device_id, sample_rate_hz, expected_channels and measured_at. No unit, no sensitivity.	MISSING
and the endpoint would not take it	POST /api/v1/ingest/raw has no field for either. It does accept rotation_speed_rpm, which the gateway also never sends, while the equipment record already holds 1480 rpm.	MISSING
so the database says	signal_unit = 'unconfirmed', unit_confirmed = false, and one sensitivity of 100 mV/g standing for all eight channels.	MISSING

Three small changes close it: send the two fields from platform_push.py, accept them on the ingest endpoint, and write them onto the sensor record. This is code, not a decision — the values are already known and already correct.

The second gap: the captures are too short to name a bearing fault

A spectrum can only separate two frequencies if they are further apart than one line, and the line spacing is one divided by the recording length. The gateway records 0.278 seconds, so the lines are 3.6 Hz apart. That is coarser than the gaps between this pump's bearing frequencies and its ordinary shaft harmonics.

Bearing frequency	Sits at	Nearest shaft harmonic	Gap	Now (3.6 Hz)	At 1 s (1.0 Hz)
FTF — the cage	9.45 Hz	1× = 24.67 Hz	15.2 Hz	separate, but only 2.6 lines wide	9.4 lines — readable
BSF — a ball	49.83 Hz	2× = 49.33 Hz	0.49 Hz	same line	still the same line
BPFO — outer ring	75.58 Hz	3× = 74.00 Hz	1.58 Hz	same line	separate
BPFI — inner ring	121.75 Hz	5× = 123.33 Hz	1.58 Hz	same line	separate

Read the fifth column: today, a mark on the outer ring of a bearing and an ordinary third shaft harmonic land on the same line of the spectrum. Looseness produces a long family of shaft harmonics, so a loose foot would look like a failing bearing. Changing LOR from 2500 to 9000 on the Senvia sensor page makes the recording one second long and settles all of it except the ball frequency, which would need about six seconds and is therefore permanently unreadable on a 50 Hz supply at this speed. The system will say so rather than guess.

Envelope analysis, the next piece of work, softens this: it demodulates the bearing's high-frequency ringing, where shaft harmonics do not appear. But its own resolution comes from the same recording length, so the change is worth making either way.

Found along the way, not on the ticket list

- **A 50 Hz line on two channels — and what it is depends on the machine** — Channels 2 and 7 carry a strong line at 50.4 Hz. On the idle captures, with the shaft stopped, it stands 93 to 97 times above the noise and can only be electrical: a stopped shaft has no orders. On running captures the same channels read 45 to 48 times above a higher floor, and there the line cannot be attributed, because twice shaft speed is 49.33 Hz and falls in the same 3.6 Hz line. Those channels also show 1×, 3× and 4× shaft, which mains cannot produce, so most of what is there is mechanical. The code already refuses to call it mains while the machine runs; this document previously did not, and that was wrong.
- **Timestamps were not increasing** — The gateway wrote one timestamp per block, so 13,888 rows carried seven distinct times and the platform rejected every file. Fixed by deriving a time per sample.
- **Credentials were in the repository** — A tracked settings file carried a live password, and .env.example held real credentials while being git-ignored. Both fixed going forward; one commit in the history still needs scrubbing, which cannot be done by one person alone.

- **One equipment record was blank and one is test data** — The blank row has been deleted — nothing referenced it. The row named 'pump' carries bearing numbers 5656 and 65421 that match nothing in the catalogue, and a speed range of 55 to 85 which looks like a load percentage typed into an rpm field. It is kept, and excluded from analysis until corrected.

What a full audit of the work found

Everything built so far was reviewed in five passes. The method was mutation testing: deliberately break one safeguard in the code and check that some test notices. A safeguard no test protects is one nobody would notice losing.

- **Two real bugs, now fixed** — The two windows used to measure whether vibration is growing could overlap, so a record that grew tenfold partway through would report almost no change. And a unit written as a capital G was refused, which would have stopped every capture being graded for a reason nobody would think to look for.
- **A test that could not fail** — The test protecting the check that twelve manufacturers' bearing figures are not averaged together asserted only inside an if-statement that the check itself populated. Switch the check off and the test passed by skipping itself. It is now pinned against the catalogue outright.
- **A test suite nobody could run** — Neither pytest nor httpx was declared, so a fresh checkout could not run the tests at all — and one script in scripts/ is named like a test and does its work on import, which took the whole run down with it. Both fixed.
- **Mutation scores before and after** — Signal units 9 of 15 to 16 of 16. Time-domain features 9 of 13 to 14 of 15. Bearing resolution 6 of 12 to 10 of 12. The remaining survivors are mutations that change no behaviour. Eighteen new tests; 251 to 269.
- **Everything else checked clean** — All 94 stored captures are well formed with no dead channels. All three migrations carry working downgrades. No secrets in any tracked file. Backend, frontend and type checks all pass.

The numbers, and where they came from

Read from the database and from git when this document was built, not copied in. Capture counts rise while the gateway runs.

Bearings in the catalogue	88,734	bearing_fault_frequencies
Captures stored	94	55 on 17 Sep, 39 on 21 Sep
Days of data	2	the roadmap wants two to four weeks before a baseline is trusted
Capture shape	8 ch × 13,888 @ 50 kHz	0.278 s, 6.85 shaft turns
Features per channel	36	was 10 at the start of Phase 0
Cost of a whole capture	0.18 s	measured on 8 channels of real-shaped data
Backend tests	269 passing	pytest, backend/tests
Frontend tests	43 passing	vitest, 2 files
Features the UI can label	10 of 36	feature_definitions still holds 10 (VIK-021)
Equipment records	2	one real machine, one test row; the blank row was deleted
Commits not pushed anywhere	40	they exist on this laptop only

The pump, as the catalogue now describes it

Cooling Water Pump 1, 1480 rpm, so the shaft turns 24.667 times a second. Drive-end bearing 6312-C3, non-drive-end 6310-C3, both resolved against the catalogue.

What	What it means	Drive end	Non-drive end
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FTF	the cage, holding the balls apart	9.45 Hz	9.40 Hz
BSF	a ball spinning with a flat on it	49.83 Hz	48.86 Hz
BPF0	a mark on the outer ring	75.58 Hz	75.18 Hz
BPFI	a mark on the inner ring	121.75 Hz	122.15 Hz

The outer and inner figures add up to 8.00, the number of balls in the bearing. That identity holding is a check that the catalogue entry is the right one. Twelve manufacturers list this bearing and their figures differ by 1.2% on the ball frequency; the spread is reported rather than averaged, because an average is a number none of them publishes.

What Phase 1 inherits

Phase 0 said nothing in Phase 1 starts until VIK-005, VIK-009 and VIK-013 are merged. Two of the three are. VIK-013, the background worker, is not, and its stated reason was that Phase 1's extra features would make uploads time out. Measured, they do not: a whole capture costs 0.18 seconds, because VIK-014's restructure absorbed the increase. So the feature tickets were safe to start, and VIK-018 and VIK-019 are done. VIK-013 still matters for the phases after this one, where scoring and baselines run on every capture.

Two Phase 1 tickets are finished: the thirteen time-domain measurements and the thirteen frequency-domain ones, taking the count from 10 to 36. VIK-020, the envelope features, is the group the roadmap calls the most important one for the demo, and it is next.